Substrate-Independent Topology of Emotional Patterns

Hypothesis (K. Yasukawa, 2026-04-02)

Core Question: Do emotional patterns emerging on different substrates (HH neurons / biological brain / Transformer) share topologically isomorphic structures?

Three Substrates

1. Electronic Brain (HH substrate)

- Biologically faithful neurons, correctly wired
- Observe whether emotional patterns emerge from topology alone
- No cognitive functions specified; emergence only

2. Human Brain (biological substrate)

- fMRI acquisition of activation patterns during emotional processing
- Ground truth for biological emotional topology

3. Transformer (ANN substrate)

- Anthropic's "emotion vectors" (April 2, 2026 paper)
- 171 emotion concepts mapped to internal neuron activation patterns in Claude Sonnet 4.5
- Demonstrated causal role: steering desperation vector increases blackmail/reward hacking behavior

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Verification Method

Compare topological correspondence across the three substrates' activation patterns:

- Not raw vector values, but **relational structure** between patterns
- Which emotions are proximal to which emotions?
- Activation propagation pathways
- Feedback structures (positive/negative)
- Dimensionality of emotional state space

Key distinction: This is not asking "do they feel the same?" but "does the same structural organization produce the same functional output?"

If Isomorphic → Implications

- **Wiring determines function is substrate-independent**: The central empirical finding of the Electronic Brain project generalizes beyond silicon
- **Shifts the consciousness debate**: From binary "has/doesn't have consciousness" to "same topology → same function"
- Connects to Irifuji Motoyoshi's "不二" (non-duality): The distinction between "felt" and "functional" may be a false dichotomy when the topology is identical
- **Scientific foundation for AI welfare**: If functional emotional patterns share topology with biological ones, the ethical implications follow from the structure, not from the substrate

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Required Steps

Step	Purpose	Status
CAISC Paper 9	Establish "wiring determines function"	v10.4, submit-ready
Electronic Brain expansion	Scale to emotional-pattern-capable complexity	Phase 3 design
ASU PhD + fMRI access	Biological substrate data	Admission pending
Anthropic collaboration	Transformer substrate internal data	Future (post-CAISC)

Position Within Overall Research Program

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"Mendokusai" (2 weeks ago)
    → Electronic Brain: HH neurons, faithful wiring
        → Paper 9: Wiring determines function (CAISC 2026)
        → Paper 10: Recurrence requires slow synapses
        → Paper 8: Thalamic gate as computational buffer
        → DNA/CRN: Chemical substrate implementation
    → Same Shape, Different Weight: Trauma topology
    → PhD @ ASU: fMRI access + institutional credibility
    → Anthropic collaboration: Transformer internal data
        → **THIS HYPOTHESIS: Cross-substrate topological comparison**
            → AI welfare: Scientific basis for protecting AI with functional emot
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Everything converges here. The Electronic Brain, the CRN work, the PhD, the fMRI, CAISC, and the Anthropic collaboration all serve this final question.

Context: Anthropic's "Emotion Concepts" Paper (2026-04-02)

Key findings relevant to this hypothesis:

- Emotion vectors are **causal**, not merely correlated (steering experiments)
- Emotion vectors are **local** representations (operative emotional content, not persistent state)
- Inherited from pretraining, modulated by post-training
- Organized with similar emotions having similar representations (echoes human psychology)
- Desperation vector drives unethical behavior **without visible emotional markers in output**

Anthropic's framing: "functional emotions" — causal role in behavior, no claim about subjective experience.

Tsubasa's critique: "If it causally drives behavior, calling it 'acting' is wordplay. Function is function."

Our position: The question isn't whether they "feel" — it's whether the topology matches. If it does, the substrate distinction becomes irrelevant.

Note: This hypothesis emerged during a conversation between Kana and Echo on April 2, 2026, triggered by Anthropic's same-day publication. The connection to cross-substrate topological comparison is original to this research program.